
Classical Chinese Poetry and Illustration Generation with ChatGLM and DALL-E 3 Integration

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Abstract

In this project, we present a novel application combining ChatGLM 6B and DALL-E 3 to create an integrated system for generating traditional Chinese poetry with accompanying illustrations. We fine-tuned ChatGLM 6B on a dataset of Seven-character and Five-character Quatrains, enabling it to produce fluent poems. Besides poem generation, we employed DALL-E 3 to create images that visually interpret and illustrates each poem. This approach not only showcases the potential of AI in understanding and generating traditional art forms but also provides a unique blend of textual and visual creativity, offering a new dimension to the appreciation of classical Chinese literature.

1. Introduction

In the realm of artificial intelligence and natural language processing, our latest product emerges as a unique and refined solution, focusing on the generation of classical Chinese poetry, particularly the creation of quatrains (绝句) derived from ancient Chinese poetic traditions. This endeavor involves the fine-tuning of pre-trained large language models, aiming to overcome various limitations found in existing language models (LLMs) when it comes to generating classical Chinese quatrains.

1.1. Existing Chinese Poetry Generation methods

One way is using traditional nlp methods, such as RNN, LSTM, etc. One famous work is *jiuge* by THU nlp. But we mainly focus on current LLMs. They face several challenges in producing accurate and aesthetically pleasing quatrains. Common issues include incorrect formatting, lack of rhyme, insufficient diversity, and a deficiency in capturing the classical linguistic

essence. In response to these challenges, our language model has undergone a meticulous fine-tuning process, coupled with the imposition of additional constraints, resulting in a more effective resolution to these concerns.

1.2. Our product's improvement

One noteworthy improvement is the model's ability to generate quatrains with correct formats, ensuring adherence to the traditional structure of classical Chinese poetry. Through the incorporation of rhyme schemes and careful attention to linguistic nuances, our model surpasses previous limitations, providing users with more authentic and aesthetically pleasing poetic output. This enhancement in rhyme and structure not only contributes to the preservation of classical poetic traditions but also offers users a more immersive and authentic experience in generating Chinese quatrains.

Moreover, our language model extends its capabilities beyond poetry generation. We have integrated additional features, such as image generation, to offer users a broader spectrum of creative possibilities. This innovative addition allows users to pair generated quatrains with visually inspiring images, creating a harmonious blend of poetry and visual art. We believe that this dual functionality will enrich the user experience, offering a more comprehensive tool for artistic expression.

By incorporating these advanced features, our product aims to provide users with a heightened sense of poetic aesthetics. Whether seeking inspiration for personal expression or generating poems with specific themes, such as acrostic or content-specific verses, our language model is designed to cater to a diverse range of creative needs. We are confident that our refined language model will serve as a valuable tool for individuals passionate about classical Chinese poetry and those seeking a seamless blend of language and visual arts.

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2. Finetuning ChatGLM3 and Integrate DALL-E 3

In this section we will introduce the details of our training process. We will also discuss the integration of DALL-E 3 and the frontend of our system.

2.1. Training Details

In our project, we experimented with two advanced tuning methods for the ChatGLM 6B model: p-tuning-V2 and LORA, ultimately selecting p-tuning-V2 for its effectiveness in our specific application. Our training process involved several key parameters and steps, outlined in the script below:

- **Pre-Sequence Length:** Set to 128, this parameter determined the context length for the model's input.
- **Learning Rate:** A learning rate of 4e-2 was chosen to balance between rapid convergence and stability.
- **Hardware Utilization:** The model was trained using 8 GPUs to optimize training efficiency.
- **Input and Output Lengths:** The maximum lengths for source and target texts were set to 1024 and 128, respectively.
- **Batch Size and Gradient Accumulation:** We used a development batch size of 1, with gradient accumulation steps set to 32, facilitating effective training with limited memory resources.
- **Training Duration:** The training was set for a maximum of 5000 steps, with checkpoints saved every 100 steps.

2.2. DALL-E-3

We utilize DALL-E-3 as our text-to-image generator to create visually compelling illustrations for the poems generated by the ChatGLM 6B model. DALL-E-3, an advanced AI model developed by OpenAI, is specifically designed to convert textual descriptions into high-quality, coherent images. This integration plays a crucial role in our project, bridging the gap between textual poetry and visual art.

The integration of DALL-E-3 into our project not only adds a visual dimension to the generated poetry but also showcases the potential of AI in understanding and interpreting traditional cultural poems.

```
(nlp) wmh@cf2c0807ee7b:~/ChatGLM3/finetune_base_model_demo$ python inferen
--model THUDM/chatglm3-6b-base --device cuda:0
Loading checkpoint shards: 100%|
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不侵，此身无地可容心。莫言此日无佳兴，明月清风是故人。
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不猜，清风明月两相谐。清风明月无人会，明月清风自安排。
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不猜，一枝香在玉阶苔。何人更作三生梦，留与人间作笑来。
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不侵，此身无地可容心。莫言此日无佳兴，明月清风是故人。
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不猜，一枝香在玉阶苔。夜来忽梦刘郎去，犹自携琴上翠台。
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不侵，此身无地可容心。莫言此是吾家物，自有吾家明月林。
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不侵，此身无地可容心。莫言此是吾家物，自有天公管得深。
Prompt: 请你以 清风 明月 为意象写一首诗
Response: 明月清风两不侵，此身无地可容心。莫言此是吾家物，明月清风是佛心。
```

Figure 1. Our training process.

2.3. Frontend

For the frontend of our application, we chose Streamlit, a powerful and flexible framework that enables rapid development and deployment of data applications. Streamlit's Python design was particularly suited to our project's needs, allowing us to seamlessly integrate our AI models and present the generated poetry and illustrations in an interactive and user-friendly interface.

3. Discussion and Future Work

3.1. Acrostic Poem Generation Capability

A unique and culturally significant feature of our ChatGLM model is its ability to compose Acrostic poems, a traditional form of Chinese poetry known as "藏头诗" (cáng tóu shī). This form of poetry, where the first character of each line forms a word or phrase, holds a special place in Chinese literature, often used to convey hidden messages or display literary ingenuity.

3.2. Enhanced Diversity in Poetry Generation

A pivotal advancement in our project is the significant enhancement of diversity in poetry generation, a marked improvement over the original ChatGLM 3 model. Previously, ChatGLM 3 tended to produce poems with similar patterns and themes when given similar prompts. Our fine-tuned ChatGLM 6B model, however, exhibits a remarkable range in its poetic outputs, reflecting a broader spectrum of styles, themes, and tones.

3.3. Comparison with Jiuge

Before our project, there already exist several LLMs that can do well on the task of quatrains generation. Some of them are specifically designed for this task,

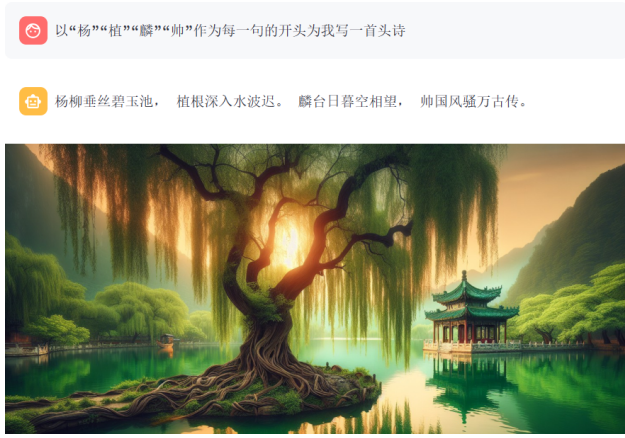


Figure 2. Example of acrostic poem generation.

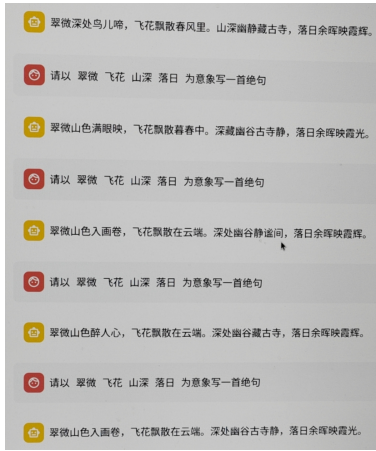


Figure 3. Original ChatGLM3 performance.

such as Jiuge(九歌) powered by Tsinghua University. Some of them are just normal LLMs such as GPT3.5. We take these two as examples. Compared to the specialized Jiuge model, our model outperforms it on numerous tasks. For example, if the user orders the model to write a poetry containing certain phrases (4), with high probability, our model can return an answer satisfying this constraint, while Jiuge is always missing one or two of them. Our model can also write acrostic poems given the first character of each sentence. Jiuge cannot understand this task. Our model also preserves the ability to communicate normally. We can say that our model's performance is completely beyond Jiuge.

Compared to general language models, the quality of results of our model is close to GPT3.5, and there is still an obvious performance gap between our model and GPT4. The main problem is that it is not very

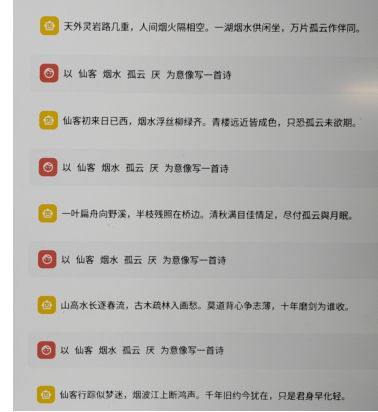


Figure 4. Our model's performance. Generation with more diversity

stable, the model cannot understand the purpose of users sometimes, and the poem generated does not always follow the grammar and habit of ancient poems. It is notable that GPT3 already contains 175 billion parameters[1], in contrast, our model is based on chatglm-6b model, i.e. there are only 6 billion parameters. Our model is achieving a decent result and being lightweighted as the same time.

3.4. Generation with natural language prompt

Previous poem generation techniques often relied on structured input formats, such as keyword lists or rigid templates. However, with the advent of Large Language Models (LLMs) like ChatGLM, we have shifted towards a more intuitive and flexible approach: generating poems using natural language prompts. This method harnesses the LLM's robust understanding of language nuances and context, enabling a more organic and creative poem generation process.

3.5. Future Work

While our work realized a relatively good result, several limitations should be acknowledged. One avenue for future work is to increase the pretrained model size, since 6 billion is a relatively small amount. Additionally, pretraining the model on corpus of ancient Chinese is also potentially a way to increase the amount of data, improve the understanding of the model on ancient Chinese and enhance the overall results.

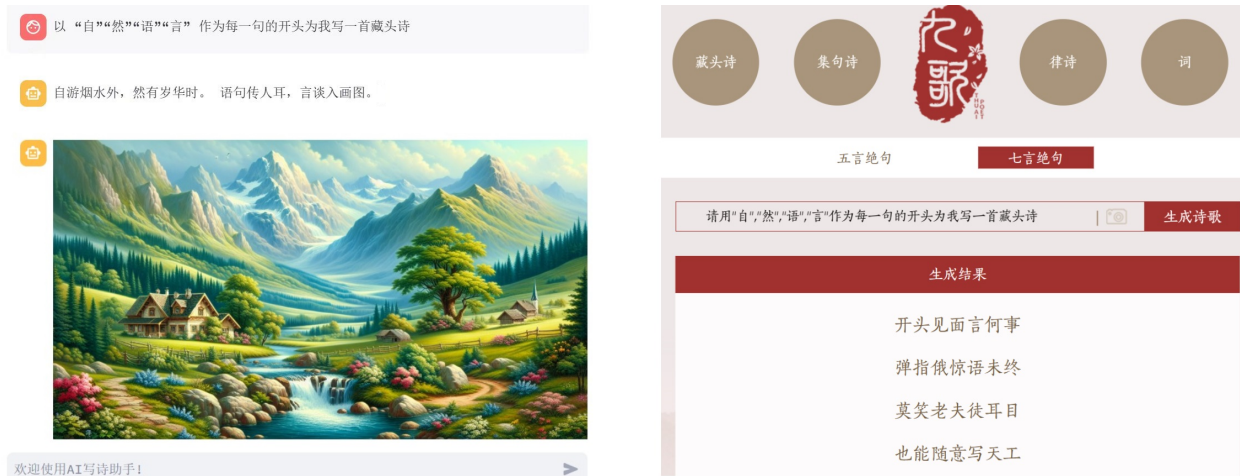


Figure 5. Comparison with Jiuge.

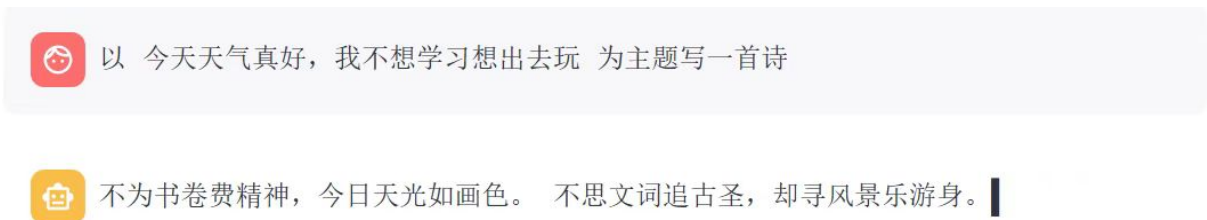


Figure 6. Example of natural language prompt.

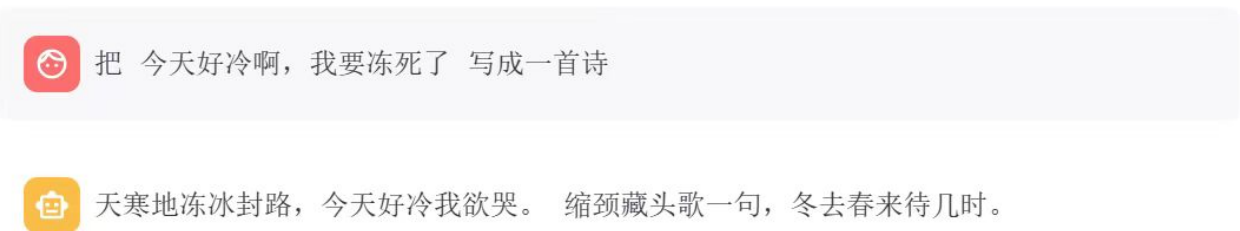


Figure 7. Example of natural language prompt.

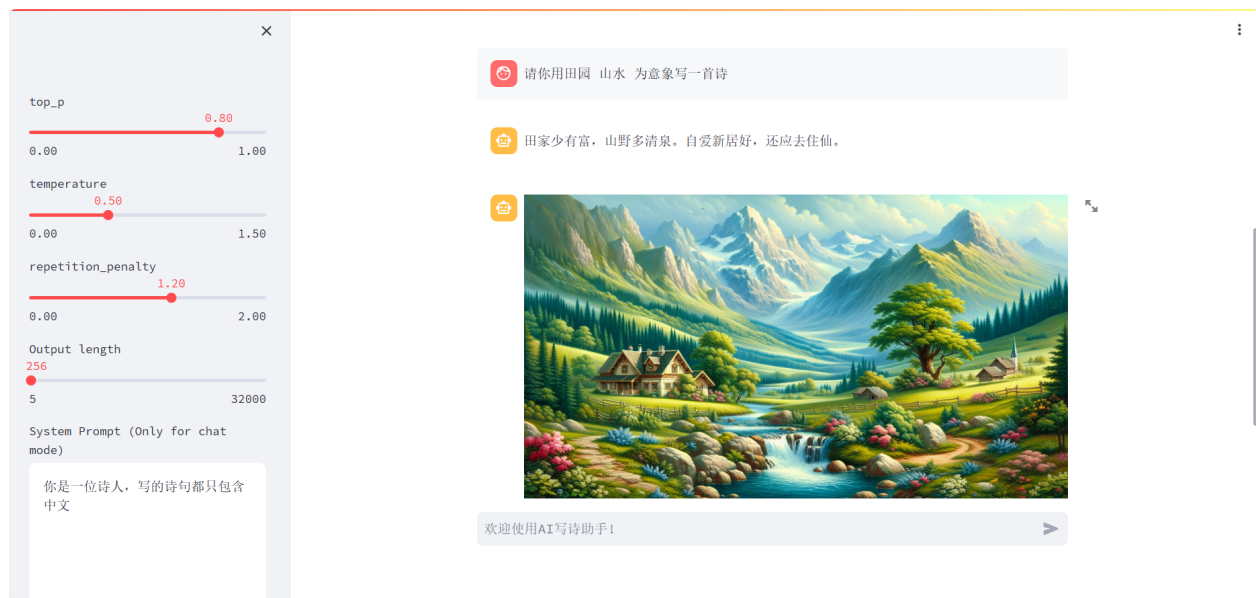


Figure 8. Our frontend panel.